

Faculty of Engineering and Technology

COMP333

Databases

Course Project Report

GrandPharm Pharmacy

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Project Description

The GrandPharm Pharmacy Management System is a database application designed to manage the pharmacy's daily operations involving medications, healthcare products, customers, suppliers, warehouses, and transactions.

The system will help organize prescription and non-prescription medicines, maintain accurate stock records, and support the handling of purchases and sales in an efficient way. The system will make it easier to check product availability, follow stock movement, and reduce mistakes in daily work.

Customers will be served through a more organized process for purchasing products and processing prescriptions, while administrators and employees will be able to manage customers, products, suppliers, warehouses, and staff records. The system will also help monitor stock levels, track batch and expiration information, and keep records of sales and purchase transactions, in addition to providing useful system reports and statistics.

Project Scope

The scope of the project includes managing customers, employees, suppliers, products, categories, warehouses, prescriptions, inventory, purchases, sales transactions, and reporting. Administrators will be able to manage product records, update stock information, manage warehouse data, and keep track of transactions and medicine availability. Employees will be able to process sales, handle prescriptions, and record purchase operations in the system.

The system will be developed using Java for the application side and MySQL for the database, according to the project requirements. The project will focus on the pharmacy's internal database operations and management functions, including stock control, warehouse records, and transaction processing.

Client Information

GrandPharm Pharmacy is the client for this project. It is a pharmacy that deals with medications, healthcare products, suppliers, stock, and daily sales transactions, which makes it suitable for a database application of this type. Since the pharmacy handles both prescription and non-prescription medicines, it requires accurate management of products, inventory, purchases, sales, and related records.

The proposed system is intended to support the pharmacy's daily work by organizing data related to products, customers, employees, suppliers, warehouses, prescriptions, purchases, and sales. GrandPharm Pharmacy will be used as the main real-life reference for identifying the system requirements and designing a database structure that matches the pharmacy's actual needs.

Contact Info

You can contact GrandPharm Pharmacy through social media pages:

Instagram: grandpharm_pharmacy

Facebook: GrandPharm

Branches

City Center, Ramallah

Al Sharafa, Bireh

System and Database Requirements

The proposed system supports managing customers, employees, suppliers, products, categories, warehouses, prescriptions, purchases, sales, and inventory. It allows users to add, update, delete, search for, and retrieve data in a clear and organized way. The system should also help in tracking medicine availability, stock quantities, batch details, expiration dates, and daily transactions, in addition to generating reports related to sales, purchases, and inventory.

From the database side, the system should keep accurate relationships between the different entities using primary keys and foreign keys. It should support the relationships needed in the project, especially for sales, purchase orders, prescriptions, and warehouse stock. The database should also support SQL queries that include joins, filtering, sorting, grouping, and aggregation so the pharmacy can get useful information and follow its operations more easily.

Entities

1. **Branch**
The **Branch** entity stores information about the pharmacy branches.

Attributes: **BID**, Name, City, Phone, Location

2. **Warehouse**
The **Warehouse** entity stores information about the warehouses or storage locations used by the pharmacy.

Attributes: **WareID**, WareName, BID, Location, Capacity

3. **Employee**
The **Employee** entity stores information about pharmacy employees.

Attributes: **EID**, Name, Role, Phone, HireDate, Salary, BID

4. **Customer**
The **Customer** entity stores information about customers.

Attributes: **CID**, Name, Phone, DOB, Gender

5. **Supplier**
The **Supplier** entity stores information about the suppliers that provide medicines and healthcare products.

Attributes: **SuppID**, Name, Phone, Email, Address

6. **Category**
The **Category** entity stores the different product categories in the system.

Attributes: **CatID** , Name, Description

7. **Product**
The **Product** entity stores information about the medicines and healthcare products available in the pharmacy.

Attributes: **ProdID**, ProdName, Price, Prescription_Requirement, CatID

8. **Inventory** (Not shown in ER)
The **Inventory** entity stores information about the quantity of each product available in each warehouse, including stock levels, and the last update date for stock records

Attributes: **WareID**, **ProdID,** QuantityAvailable, MinStock, LastUpdated

9. Sale
The **Sale** entity stores information about sales transactions in the pharmacy.

Attributes: **SID**, Date,Time, TotalAmount, CID, BID, EID, PayMethodID

10. **PaymentMethod**
The **PaymentMethod** entity stores information about the available payment methods used in sales transactions.

Attributes: **PayMethodID**, MethodName, Description

11. **PurchaseOrder**
The **PurchaseOrder** entity stores information about purchase transactions made by the pharmacy to suppliers.

Attributes: **PurchID**,Date, TotalCost, SuppID, BID, EID

12. **Prescription**
The **Prescription** entity stores information about prescriptions submitted by customers.

Attributes: **PrescID**, Date, Doc_Name, CID, BID, EID

13. **PrescriptionItem** (Not shown in ER)
The **PrescriptionItem** entity stores information about the products included in each prescription (Not shown in ER)

Attributes: **PrescID**, **ProdID**, Dosage, Duration, Instructions

14. **SaleItem** (Not shown in ER)
The **SaleItem** entity stores the products included in each sale.

Attributes: **SID**, **ProdID**, Quantity, UnitPrice

15. **PurchaseItem** (Not shown in ER)
The **PurchaseItem** entity stores the products included in each purchase order.
Attributes: **PurchID**, **ProdID**, Quantity, UnitCost

16. **Batch**

The **Batch** entity stores information about product batches kept in the warehouse.
Attributes: **WareID** , **BNum**, ProdID, ManufacturingDate, ExpDate, QuantityReceived, QuantityRemaining

17. **SupplierProduct** (Not shown in ER)

The **SupplierProduct** entity stores information about the products that each supplier can provide to the pharmacy, including supplier price, delivery time, and availability status.
Attributes: **SuppID**, **ProdID**, SupplierPrice, DelivTime, Availability

Relations

1) **Branch-Employee (1-M):**

Each branch can employ many employees while each employee works at exactly one branch

2) **Branch-Sale (1-M):**

Each branch can process many sales while each sale takes place at exactly one branch

3) **Branch-PurchaseOrder (1-M):**

Each branch can create many purchase orders while each purchase order belongs to exactly one branch

4) **Branch-Prescription (1-M):**

Each branch can handle many prescriptions while each prescription is submitted at exactly one branch

5) **Customer-Sale (1-M):**

Each customer can make many sales while each sale is associated with exactly one customer

6) **Customer-Prescription (1-M):**

Each customer can submit many prescriptions while each prescription belongs to exactly one customer

7) **Supplier-PurchaseOrder (1-M):**

Each supplier can provide many purchase orders while each purchase order is made to exactly one supplier

8) **Category-Product (1-M):**

Each category can include many products while each product belongs to exactly one category

9) E**mployee-Sale (1-M):**

Each employee can handle many sales while each sale is handled by exactly one employee

10) **Employee-PurchaseOrder (1-M):**

Each employee can manage many purchase orders while each purchase order is handled by exactly one employee

11) **Employee-Prescription (1-M):**

Each employee can process many prescriptions while each prescription is processed by exactly one employee

12) **Sale-Product (M-M):**

Each sale can contain many products and each product can appear in many sales
this appears in the **SaleItem** Entity.

13) **PurchaseOrder-Product (M-M):**

Each purchase order can include many products and each product can appear in many purchase orders which is represented in the **PurchaseItem** Entity

14) **Branch-Warehouse (1-M):**

Each branch can have many warehouses while each warehouse belongs to exactly one branch

15) **Warehouse-Batch (1-M):**

Each warehouse can contain many batches while each batch belongs to exactly one warehouse

16) **Product-Batch (1-M):**

Each product can have many batches while each batch belongs to exactly one product

17) Warehouse-Product / Inventory (M-M)

Each warehouse can store many products, and each product can be stored in many warehouses. This relationship is represented by the **Inventory** entity.

18) PaymentMethod-Sale (1-M)

Each payment method can be used in many sales, while each sale uses exactly one payment method.

19) Prescription-Product (M-M)

Each prescription can include many products, and each product can appear in many prescriptions. This relationship is represented in the **PrescriptionItem** entity.

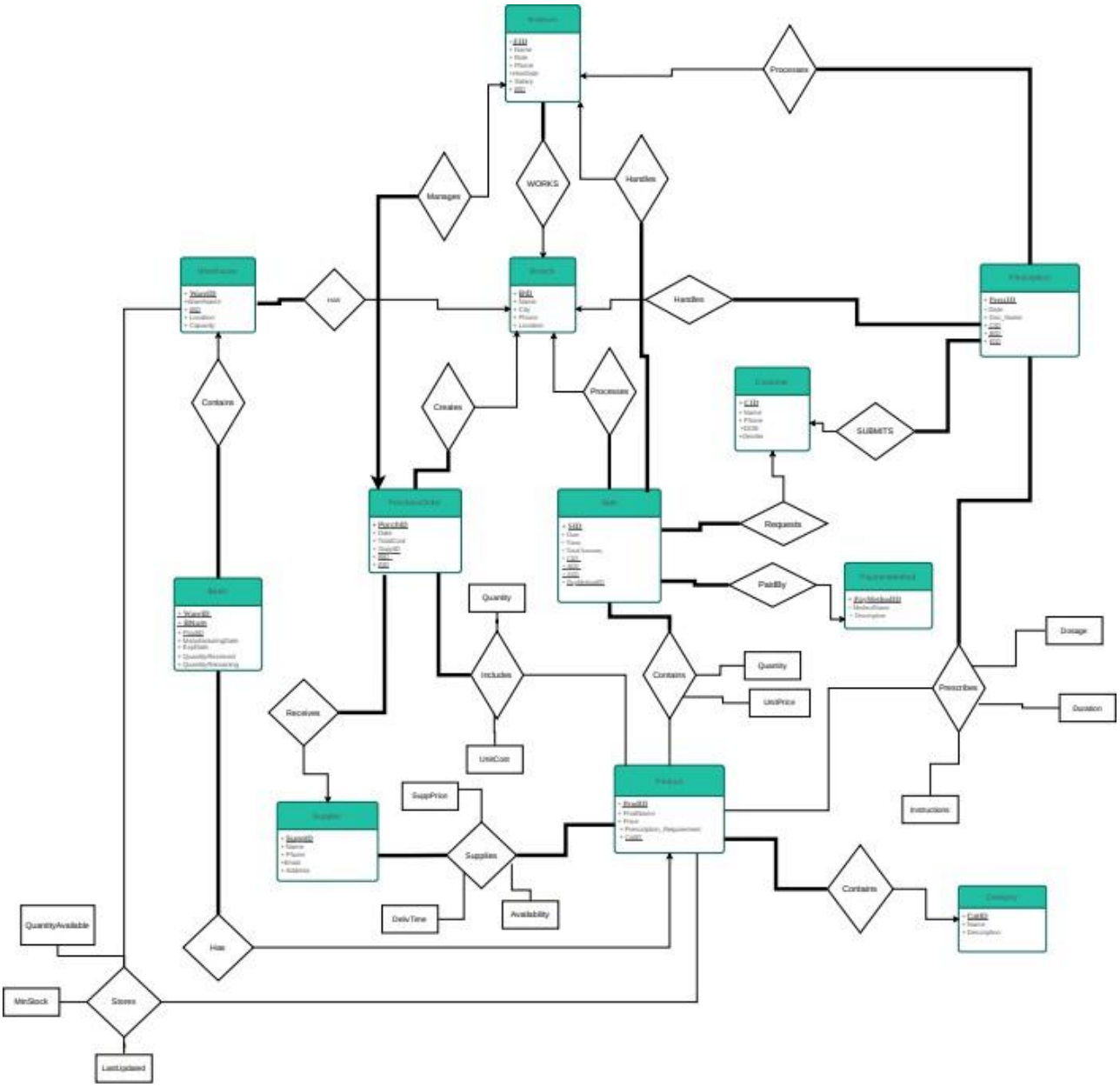
20) Supplier-Product (M-M)

Each supplier can provide many products, and each product can be provided by many suppliers. This relationship is represented by the **SupplierProduct** entity

Sample Queries

1. Retrieve all employees with their branch names.
2. Retrieve all customers sorted by name.
3. Retrieve all products with their categories.
4. Retrieve all products with their category names and prices, sorted by category then product name
5. Retrieve all suppliers and the number of purchase orders made to each supplier
6. Retrieve all sales with customer name, branch name, and employee name
7. Retrieve the total quantity sold for each product
8. Retrieve all employees and the number of sales handled by each employee
9. Retrieve all batches that will expire before a specific date
10. Retrieve all customers who made sales above a specific total amount
11. Retrieve all products that were never included in any sale
12. Retrieve the total purchase cost for each supplier
13. Retrieve total sales revenue generated by each branch, ordered by highest revenue.
14. Retrieve the top-selling products by total quantity sold.
15. Retrieve the number of sales made per payment method
16. Retrieve daily sales revenue trend, grouped by sale date
17. Retrieve total inventory quantity available per product category
18. Retrieve total revenue generated by each employee, ordered by highest revenue
19. Retrieve total amount spent on purchase orders per supplier, ordered by highest spending
20. Retrieve the number of prescriptions issued per month, formatted as year-month

ER Diagram



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ER Diagram Explanation

The diagram shows the main workflow of GrandPharm pharmacy model, as each branch is connected to the employees, warehouses, sales, purchase orders, and prescriptions that belong to it. Products are organized under categories, stored in warehouses, received through purchase orders from suppliers, sold through sales, and linked to prescriptions when needed. Batches are also connected to both products and warehouses so the system can track received quantities, remaining quantities, manufacturing dates, and expiration dates, all of these relations and entities were already mentioned in the entities and relations part.

Some many-to-many relationships are shown as relationships with attributes instead of separate entity boxes in the ER diagram. This includes Sale–Product, PurchaseOrder–Product, Warehouse–Product, Prescription–Product, and Supplier–Product. These relationships contain important attributes such as quantity, unit price, dosage, stock level, supplier price, and availability. Later, when converting the ER diagram into the relational schema, these relationships will be converted into separate tables such as SaleItem, PurchaseItem, Inventory, PrescriptionItem, and SupplierProduct.

The cardinality of the relations was already specified in the relations section, so below is only a list explaining the total participations of the entities in some relations

1) **Employee in Branch–Employee:**

Each employee must work at one branch, so Employee has total participation.

2) **Warehouse in Branch–Warehouse:**

Each warehouse must belong to one branch, so Warehouse has total participation.

3) **Sale in Branch–Sale:**

Each sale must take place at one branch, so Sale has total participation.

4) **PurchaseOrder in Branch–PurchaseOrder:**

Each purchase order must be created by one branch, so PurchaseOrder has total participation.

5) **Prescription in Branch–Prescription:**

Each prescription must be handled by one branch, so Prescription has total participation.

6) **Sale in Customer–Sale:**

Each sale must be linked to one customer, so Sale has total participation.

7) **Prescription in Customer–Prescription:**

Each prescription must belong to one customer, so Prescription has total participation.

8) **PurchaseOrder in Supplier–PurchaseOrder:**

Each purchase order must be linked to one supplier, so PurchaseOrder has total participation.

9) **Product in Category–Product:**

Each product must belong to one category, so Product has total participation.

10) **Sale in Employee–Sale:**

Each sale must be handled by one employee, so Sale has total participation.

11) **PurchaseOrder in Employee–PurchaseOrder:**

Each purchase order must be managed by one employee, so PurchaseOrder has total participation.

12) **Prescription in Employee–Prescription:**

Each prescription must be processed by one employee, so Prescription has total participation.

13) **Batch in Warehouse–Batch:**

Each batch must be stored in one warehouse, so Batch has total participation.

14 **Batch in Product–Batch:**

Each batch must belong to one product, so Batch has total participation.

15 **Sale in PaymentMethod–Sale:**

Each sale must use one payment method, so Sale has total participation.

Normalization

The database was normalized to Third Normal Form (3NF) to reduce data redundancy and improve data consistency. First, each table was designed to store information about a single entity, such as Customer, Employee, Product, Supplier, or Branch. This ensures that all attributes in a table describe only that entity.

Many-to-many relationships were separated into additional tables. For example, SaleItem was created to represent the relationship between Sale and Product, PurchaseItem was created for PurchaseOrder and Product, and PrescriptionItem was created for Prescription and Product. This prevents duplicate data and allows the system to store multiple products within the same transaction.

In addition, non-key attributes depend only on the primary key of their table. Information such as customer details, supplier details, and branch details is stored in separate tables and linked using foreign keys. As a result, the design avoids dependencies and minimizes updates.